

Comparative Performance and Computational Complexity Analysis of Pre-Trained Deep Learning Models for Citrus Fruit Disease Classification

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ABSTRACT

Citrus fruit diseases pose a serious problem for agricultural productivity and if they are not detected early enough, the losses can be very difficult to recover. This paper uses transfer learning with four pre-trained CNN models (MobileNetV2, InceptionV3, ResNet50, and VGG19) to classify citrus diseases and also analyzes their computational costs for practical deployment. A dataset of 1,463 RGB images covering four classes (canker, black spot, greening, and healthy) was used for training and evaluation. Among all models, MobileNetV2 achieved the best results with 96.77% accuracy, 98.54% F1-score, a training time of only 406.29 s, and just 9.82 MB of memory. InceptionV3 came close in accuracy at 96.59% but needed more memory (22.81 MB). ResNet50 did not perform well under frozen-base fine-tuning, reaching only 58.21% F1-score. VGG19 took the longest to train (3360.14 s) and used the most memory (118.59 MB). The results show that lightweight architectures like MobileNetV2 are well suited for building mobile and edge-based disease detection tools in the field.

General Terms

Algorithms, Experimentation, Performance

Keywords

Transfer Learning, Convolutional Neural Networks, Citrus Disease Classification, Performance Metrics, Computational Complexity, MobileNetV2

1. INTRODUCTION

Agriculture is one of the most important sectors in Bangladesh as a large part of the economy depends on it directly. Citrus fruit production has seen rapid growth over the past five decades, going from 19,163 tonnes in 1973 to 175,325 tonnes in 2022 [1]. But citrus orchards in regions like Sylhet, Moulvibazar, and Habiganj face persistent problems from diseases such as canker, black spot, scab, and greening (also called Huanglongbing) [2, 3]. When these infections are not caught early, the fruit yield drops and the cosmetic damage lowers the price at market [4, 5, 3]. Traditionally, disease checking has been done manually by agricultural officers or the farmers themselves, which is slow and not always accurate, especially when there are large areas of orchards to cover [6, 7]. Computer vision gives an automated alternative where RGB images taken in the field can be processed directly [8, 9]. CNNs have proven to be quite good at picking up visual features from plant images, and architectures like AlexNet, VGG16, VGG19, InceptionV3, ResNet, and DenseNet have shown high accuracy on botanical image datasets [2].

1.1 Context and Background

In Bangladesh, a very large portion of rural households earn their living from seasonal crop and fruit harvests [10, 11]. About 39,678 acres of land are allocated for fruit and crop production across the country [12, 10]. Since field conditions can change quickly due to weather shifts and pest outbreaks, getting a fast diagnosis

is important so that farmers can spray the right pesticides without wasting chemicals on unaffected areas [3, 9].

1.2 Research Motivation

Several studies have already shown that deep learning models can classify plant diseases with high accuracy [5, 2, 11]. However, many of these works use heavy architectures like AlexNet or VGG19, which need a lot of GPU memory and take long to train [2]. In practical agricultural settings, the software often has to run on smartphones, small microcontrollers, or drones that have limited battery and memory. Not many existing studies actually measure things like how long the model takes to run, how much storage it needs, or how much RAM it consumes alongside the accuracy numbers. This paper tries to fill that gap by testing both classification metrics and hardware overhead across four pre-trained models.

1.3 Objectives and Scope

The objectives of this work are:

- Fine-tuning four pre-trained CNN models (MobileNetV2, InceptionV3, ResNet50, and VGG19) using transfer learning on 1,463 citrus fruit images across four distinct conditions.
- Measuring classification metrics (accuracy, precision, recall, F1-score) and hardware-oriented parameters (training time, test inference latency, parameter count, RAM usage).
- Identifying the most viable model architecture for deployment on portable, battery-powered agricultural hardware.

1.4 Significance of Research

This research is done to provide concrete engineering benchmarks that link diagnostic accuracy directly to embedded hardware constraints in citrus disease detection. The data can serve as reference points for building field-ready mobile apps and drone-based scouting systems.

The structure of this paper is as follows. Section 2 presents the literature review. Section 3 discusses the methodology and model configurations. Section 4 covers the performance evaluation and complexity benchmarks. Section 5 presents the conclusions and future directions.

2. LITERATURE REVIEW

The people of Bangladesh heavily depend on agriculture. According to available statistics, about 87% of the rural population directly or indirectly depends on agricultural activities [13]. Because of this, early detection of plant diseases using automated methods has attracted a lot of research attention.

Ahmed and Muqit [5] designed a deep neural network pipeline for detecting citrus fruit and leaf infections. They used 2,293 images from PlantVillage and got an accuracy of 94.55%. Balafas et al. [8] did a comparative study between traditional machine learning models (SVM, Random Forest, SGD) and deeper networks (InceptionV3, VGG16, VGG19). Their results showed that deep features handle noisy backgrounds and different lighting conditions better than shallow hand-crafted descriptors.

For identifying diseases in citrus, Elaraby et al. [2] used a deep learning approach and achieved 94.0% accuracy with AlexNet and VGG19 backbones. Sujatha et al. [6] tested deep architectures for mobile plant pathology tools and pointed out the need for making diagnostics accessible at the field level. Hasan et al. [10, 22] applied transfer learning on regional crop datasets from Bangladesh, and

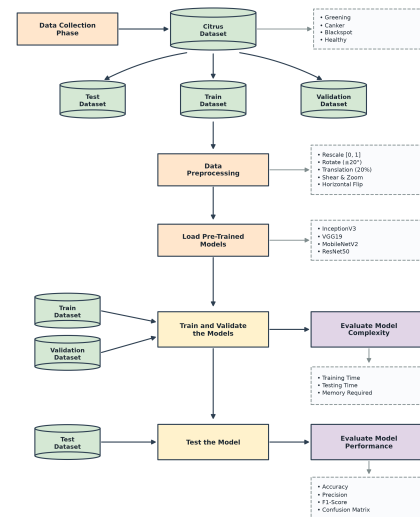


Fig. 1. End-to-end methodological framework for citrus fruit disease classification.

they found that using pre-trained weights from ImageNet helped a lot with feature convergence on small target datasets. Table 1 gives a summary of recent related studies.

3. PROPOSED METHODOLOGY

The dataset was carefully collected and preprocessed to make it suitable for image processing. TensorFlow and Keras were used as the main framework for building and training the models. Four CNN architectures were selected for comparison: MobileNetV2, InceptionV3, ResNet50, and VGG19. The detailed workflow is shown in Figure 1.

3.1 Dataset Overview and Preparation

A dataset of 1,463 high-resolution RGB images has been collected from PlantVillage and Kaggle repositories. These images are sorted into four classes:

- **Canker:** Raised, rough lesions on the fruit surface caused by the bacterium *Xanthomonas citri*.
- **Black Spot:** Dark circular spots caused by the fungus *Phyllosticta citricarpa*.
- **Greening (HLB):** Asymmetric yellowing of fruit and leaves, caused by the vector-borne bacterium *Candidatus Liberibacter*.
- **Healthy:** Normal fruit samples without any visible disease symptoms.

3.1.1 Class Distribution and Partitioning. The images were organized into folders by class and the counts are fairly balanced: 358 healthy, 387 greening, 362 black spot, and 356 canker. Figure 2 shows this distribution and Figure 3 displays some sample images from each class.

The dataset was split into 80% for training (1,170 images), 10% for validation (146 images), and 10% for testing (147 images). This split is shown in Figure 4. The test set was kept completely separate and only used after training was finished.

3.1.2 Data Pre-Processing and Augmentation. To prevent overfitting and make the models work better with different camera

Table 1. Summary of Related Literature on Citrus and Plant Disease Detection

No.	Author	Year	Class Types	Dataset Size	Classifier Architecture	Accuracy
1	R. Sujatha et al. [6]	2021	Black spot, Canker, Greening, Melanose, Healthy	609 images	LSVM, SGD, RF, InceptionV3, VGG16, VGG19	89.5%
2	Khattak et al. [12]	2021	Black spot, Canker, Scab, Greening, Melanose, Healthy	2293 (Fruits+Leaves)	Custom CNN + Softmax	94.55%
3	A. Elaraby et al. [2]	2022	Anthracoise, Black spot, Canker, Scab, Greening, Melanose	759 (Fruits+Leaves)	AlexNet, VGG19	94.0%
4	S. Perez et al. [14]	2023	Multiple plant pathological conditions	Plant leaves	Lightweight CNN + Softmax	99.0%
5	Dehuan Luo et al. [15]	2023	Anthracoise, Canker, Melanosis, Scab, Brown Spot, Leaf Miner	3202 images	Light-SA YOLOv8	92.6%
6	Madhusudan G. et al. [16]	2023	Black spot, Melanose, Canker, Greening, Healthy	609 RGB images	CNN, ResNet152V2, DenseNet121, InceptionResNetV2	98.0%
7	Poonam Dhiman et al. [17]	2023	HLB, Black spot, Melanose, Canker, Scab	2950 images	CNN + LSTM	98.25%
8	W. Gómez-Flores et al. [18]	2024	HLB, Greasy spot, Deficiencies, Citrus Mite, Red scale	953 images	DNN, Transfer Learning	Not reported
9	Nouman Butt et al. [19]	2024	Canker, Citrus scab, Black spot	2184 images	SVM, KNN	99.6%
10	J. Shermila et al. [20]	2024	Fruit blight, Greening, Scab, Melanoses	759 images	Custom CNN + LSTM	96.0%
11	A.K. Saini et al. [21]	2024	Anthracoise, Melanose, Brown Spot	759 images	SADCNN-ORBM	99.76%

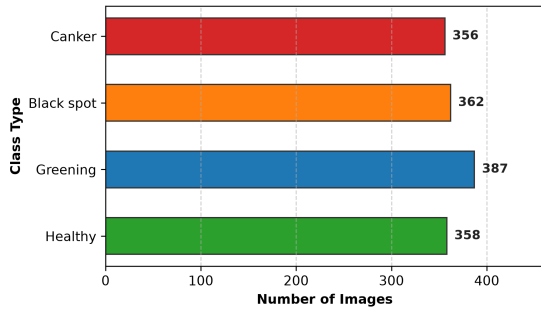


Fig. 2. Distribution of image samples across the four citrus fruit classes.



Fig. 3. Representative sample images from the experimental citrus dataset.

angles and lighting conditions, the following augmentations were applied during training:

- Rescaling pixel values into the range $[0, 1]$.
- Random rotation up to $\pm 20^\circ$.
- Horizontal and vertical translation up to 20%.
- Random shearing of images.
- Random zoom up to 20%.
- Random horizontal flipping.

All images were resized to $224 \times 224 \times 3$ pixels and loaded in mini-batches of 32.

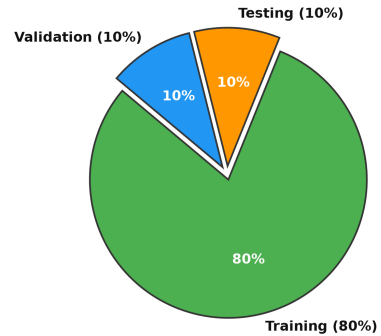


Fig. 4. Dataset partitioning into training (80%), validation (10%), and test (10%) subsets.

3.2 Transfer Learning Framework

In deep learning, transfer learning is a method where a model trained for one task is used as a starting point to solve another related problem [23, 22]. This approach speeds up learning and gives better results even on small datasets because the model already knows how to extract useful features from images. Transfer learning allows using pre-trained models which show good accuracy even on medium-sized datasets. Figure 5 shows how this process works.

Four pre-trained models have been used in this research. The working process of each model is almost the same: after loading the base model, custom layers are added on top for the classification task. The details are given below:

- **VGG19:** VGG19 is known for its simple and consistent architecture. It has 16 convolutional layers and 3 dense layers, all using 3×3 filters [24, 25, 26]. The base model was pre-trained on ImageNet and its layers were frozen during training so that the learned features are preserved. On top of the base, a 40% Dropout layer was added to reduce overfitting

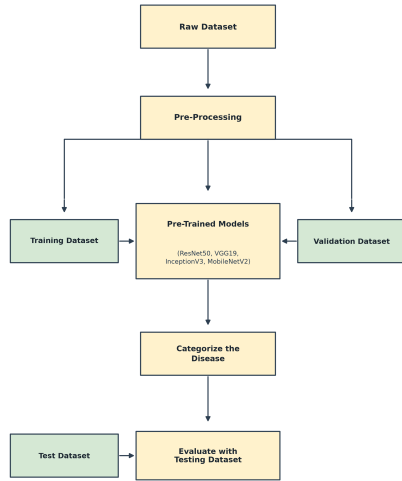


Fig. 5. Flowchart of the transfer learning fine-tuning pipeline.

[27, 28, 29], followed by a Flatten layer [30], a Dense layer with 512 ReLU units, another 20% Dropout layer, and finally a 4-unit Softmax output layer for the classification [31]. Table 2 shows the layer details.

Table 2. Layer Architecture and Parameter Details of Fine-Tuned VGG19

Layer (Type)	Output Shape	Parameters
VGG19 (Functional Base)	(None, 7, 7, 512)	20,024,384
dropout_1 (Dropout 40%)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
dense_1 (Dense, ReLU)	(None, 512)	12,845,056
dropout_2 (Dropout 20%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		32,871,492
Trainable Parameters:		12,847,108
Non-Trainable Parameters:		20,024,384

- **MobileNetV2:** MobileNetV2 uses depthwise separable convolutions and inverted residual blocks which make it much lighter than other models [32]. By freezing each of the base model layers, it keeps the features learned from ImageNet while only the final layers are customized for this task. The same dropout and dense layer setup was used on top. Table 3 shows the architecture.

Table 3. Layer Architecture and Parameter Details of Fine-Tuned MobileNetV2

Layer (Type)	Output Shape	Parameters
MobileNetV2 (Base)	(None, 7, 7, 1280)	2,257,984
dropout_1 (Dropout 40%)	(None, 7, 7, 1280)	0
flatten (Flatten)	(None, 62720)	0
dense_1 (Dense, ReLU)	(None, 512)	32,085,760
dropout_2 (Dropout 20%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		34,345,796
Trainable Parameters:		32,087,812
Non-Trainable Parameters:		2,257,984

- **InceptionV3:** InceptionV3 is a CNN architecture designed for high-performance image classification. It uses multi-scale inception modules with factorized convolutions. The pre-trained model from ImageNet was loaded without its top classification layers, and the output was fed to a Flatten layer, a Dense layer with 512 ReLU units, a 20% Dropout, and a 4-unit Softmax layer. Table 4 gives the layer summary.

Table 4. Layer Architecture and Parameter Details of Fine-Tuned InceptionV3

Layer (Type)	Output Shape	Parameters
InceptionV3 (Base)	(None, 5, 5, 2048)	21,802,784
flatten (Flatten)	(None, 51200)	0
dense_1 (Dense, ReLU)	(None, 512)	26,214,912
dropout_2 (Dropout 20%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		48,019,748
Trainable Parameters:		26,216,964
Non-Trainable Parameters:		21,802,784

- **ResNet50:** ResNet50 uses residual shortcut connections that allow gradients to pass directly through the network, which makes it easier to train very deep models. The base model was kept non-trainable to prevent overfitting on the small dataset. The top pooling layer was replaced with a Flatten layer, followed by a Dense layer with 512 units, a 50% Dropout, and a 4-class Softmax output. Adam optimizer was used with a learning rate of 0.0001. Table 5 shows the layer parameters.

Table 5. Layer Architecture and Parameter Details of Fine-Tuned ResNet50

Layer (Type)	Output Shape	Parameters
ResNet50 (Base)	(None, 5, 5, 2048)	23,587,712
flatten (Flatten)	(None, 100352)	0
dense_1 (Dense, ReLU)	(None, 512)	51,380,736
dropout_2 (Dropout 50%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		74,970,500
Trainable Parameters:		51,382,788
Non-Trainable Parameters:		23,587,712

3.2.1 Framework Implementation. All experiments were done in Python using TensorFlow and Keras. TensorFlow is a well-known open-source machine learning framework developed by the Google Brain Team. It provides a good environment for building and deploying machine learning models, and works well on CPUs, GPUs, and TPUs. The end-to-end framework, from data loading to GPU training to model export for mobile deployment (TensorFlow Lite, TensorFlow Serving), is shown in Figure 6.

3.3 Optimization and Training Configuration

All four models were trained using the Adam optimizer, which is an adaptive learning rate algorithm that combines methods from both AdaGrad and RMSProp. The learning rate was set to $\eta = 0.0001$. The loss function used was categorical cross-entropy, which is suitable for multi-class classification problems:

$$\mathcal{L}_{CCE} = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (1)$$

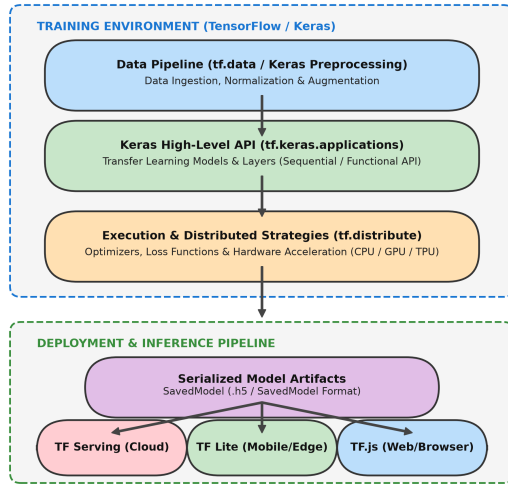


Fig. 6. Framework architecture covering training, serialization, and deployment pipelines.

where N is the mini-batch size, $C = 4$ is the number of classes, $y_{i,c} \in \{0, 1\}$ is the true label, and $\hat{y}_{i,c}$ is the predicted probability for class c . The models were trained for 50 epochs and the best weights were saved based on validation loss.

3.4 Evaluation Metrics

3.4.1 Performance Metrics. For evaluating how well the models performed, some standard parameters were used. The metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

where TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative.

3.4.2 Computational Complexity Metrics. The complexity analysis of the models was done by measuring four things: total training time in seconds, inference time on the 147 test images in seconds, trainable and non-trainable parameter count, and how much memory each model uses in MB.

4. RESULT AND DISCUSSION

This section gives a detailed discussion of the results from the experiments. The models are compared using both performance analysis and complexity analysis.

4.1 Performance Evaluation

The performance analysis was done using precision, recall, and F1-score on the 147-image test set. In this case, MobileNetV2

showed the highest precision at 98.56%, recall at 98.54%, and F1-score at 98.54%. InceptionV3 was close behind with 97.21% precision, 97.30% recall, and 97.19% F1-score. VGG19 got 84.82% precision and 81.17% F1-score. On the other hand, ResNet50 showed the lowest performance in all metrics, with only 58.21% F1-score. Table 6 presents these results.

Table 6. Performance Comparison Across Fine-Tuned Models on the Unseen Test Set

Model Name	MobileNetV2	InceptionV3	ResNet50	VGG19
Precision (%)	98.56	97.21	66.66	84.82
Recall (%)	98.54	97.30	62.02	81.31
F1-Score (%)	98.54	97.19	58.21	81.17

4.1.1 Accuracy and Loss Analysis. Each model's accuracy and loss during training, validation, and testing have been compared. High accuracy means the model is learning the data well, while low loss means good performance. If the training loss is very low but the validation loss stays high, it can indicate overfitting where the model is learning noise instead of real patterns. The training accuracy of MobileNetV2 and InceptionV3 is better than the other models, and this remains the same for validation and test accuracy. MobileNetV2 produced the lowest test loss at 5.29%. ResNet50 had the highest loss value at 76.16%, which suggests that the residual blocks may need deeper layer unfreezing or a different learning rate to work well on this particular dataset. Table 7 presents these numbers.

Table 7. Accuracy and Loss Metrics Across Training, Validation, and Test Phases

Model	Training		Validation		Testing	
	Acc. (%)	Loss (%)	Acc. (%)	Loss (%)	Acc. (%)	Loss (%)
MobileNetV2	95.80	10.97	94.55	12.80	97.29	5.29
InceptionV3	95.89	10.60	97.95	6.72	96.59	7.44
ResNet50	57.14	96.91	61.90	83.92	67.34	76.16
VGG19	75.10	60.10	87.75	32.46	88.44	37.76

4.1.2 Confusion Matrix Analysis. The confusion matrix gives an overview of how each model classified the test samples, which helps in seeing where the models made mistakes. Figure 7 shows the confusion matrices for all four models on the 147 test images. MobileNetV2 and InceptionV3 classified every greening sample correctly and made very few errors on the other classes. ResNet50 had noticeable misclassifications between healthy and diseased fruit.

4.2 Computational Complexity Analysis

The complexity analysis is an important factor in evaluating these models, especially for deployment on devices with limited resources. Training time, inference time, and memory usage were measured for each model. Table 8 compares these values.

Table 8. Quantitative Computational Complexity Metrics Across Models

Metric	MobileNetV2	InceptionV3	ResNet50	VGG19
Training Time (s)	406.29	717.00	1040.00	3360.14
Inference Latency (s)	83.67	85.73	84.11	80.34
Memory Footprint (MB)	9.82	22.81	34.28	118.59

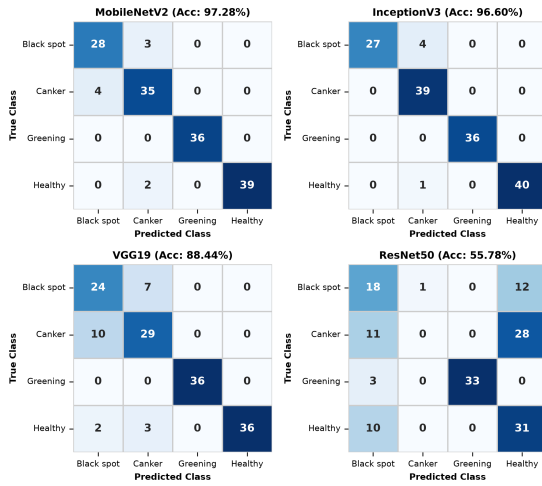


Fig. 7. Multi-class confusion matrices across all four fine-tuned CNN architectures on the test partition.

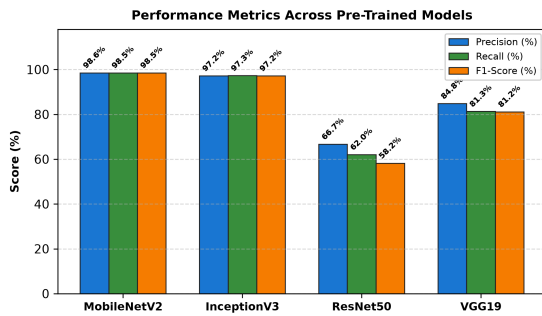


Fig. 8. Comparative diagnostic metrics across the four pre-trained architectures.

MobileNetV2 is the most efficient model with the shortest training time of 406.29 s and the lowest memory usage of 9.82 MB. Compared to VGG19, which took 3360.14 s to train and used 118.59 MB of memory, MobileNetV2 reduced training time by about 88% and memory usage by around 91.7%. The inference times were relatively close across all four models, ranging between 80.34 s and 85.73 s for the 147 test images, with VGG19 slightly faster at 80.34 s. Figure 8 shows a graphical comparison of all the metrics.

4.3 Comparative Discussion

This section gives a comparative discussion of the models used for citrus fruit disease diagnosis based on their performance and complexity metrics. In this work, the goal was to find the best model by looking at both accuracy and computational cost. The performance was evaluated using precision, recall, F1-score, and accuracy.

MobileNetV2 and InceptionV3 both had high training accuracy and low loss, showing good generalization. ResNet50 had much lower accuracy, which indicates that it did not learn the features well with frozen base layers. VGG19 showed decent results but was very heavy in terms of computation and memory.

After comparing all the results, MobileNetV2 turned out to be the best choice because it gives the highest accuracy, uses the least memory, trains the fastest, and is lightweight enough to be deployed on smartphones, portable scanners, and drone cameras for real-time disease monitoring in the field.

5. CONCLUSION

This research highlights the importance of using transfer learning with pre-trained models for detecting citrus fruit diseases. Four models were evaluated: MobileNetV2, InceptionV3, ResNet50, and VGG19, on a dataset of 1,463 images. The study concluded that MobileNetV2 is the best model among the four, giving the highest accuracy at 96.77%, an F1-score of 98.54%, a training time of 406.29 s, and only 9.82 MB of memory usage. InceptionV3 finished close behind with strong numbers, but ResNet50 was far back in terms of all metrics. VGG19 gave decent accuracy but was too heavy in computation. In the end, MobileNetV2 stands out by hitting both high accuracy and efficiency, making it the most suitable option for real-world deployment.

5.1 Limitations

This study has some limitations. The main one is that the dataset has only 1,463 images, which is relatively small for training deep learning models. Also, the images come from only two sources (PlantVillage and Kaggle), so the models were not tested under real field conditions with variable lighting, partial occlusions, or overlapping symptoms that occur in actual orchards.

5.2 Future Scope

Future work can explore several directions:

- Testing bio-inspired optimization algorithms like Grey Wolf Optimizer (GWO) or Particle Swarm Optimization (PSO) for automated hyperparameter tuning.
- Building hybrid models that combine the strengths of different architectures to get better accuracy and robustness.
- Adding explainable AI techniques such as Grad-CAM to highlight the disease regions in the image, which would help agricultural workers understand and trust the model predictions.

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